Supplement for "Informativeness under Model Uncertainty: Shadow Prices and Ridge Penalties"

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S.1 Economic interpretation of shadow prices

This subsection provides a detailed discussion of the economic interpretation of shadow prices introduced in Section 2.

Global shadow price. To clarify the interpretation of shadow prices, it is useful to note that equivalent formulations of the constrained optimization problem lead to different sign conventions for the associated Lagrange multiplier. We begin by considering the population Lagrangian associated with a constraint

$$\mathcal{L}(\theta, \lambda) = \phi(\theta) + \lambda\big(c_0 - g(\theta)'\Sigma^{-1}g(\theta)\big),$$

where λ is the Lagrange multiplier associated with the misspecification restriction and is determined endogenously, and depends on the tolerance parameter c. In this formulation,

$$\frac{\partial \mathcal{L}}{\partial c_0} = \lambda.$$

Alternatively, the equivalent constraint is written as

$$\mathcal{L}(\theta, \lambda) = \phi(\theta) + \lambda\big(g(\theta)'\Sigma^{-1}g(\theta) - c_0\big).$$

In this case,

$$\frac{\partial \mathcal{L}}{\partial c_0} = -\lambda.$$

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Note that, although the two constraint formulations are mathematically equivalent, the economic interpretation of λ depends on the chosen normalization. When the constraint is written as $c_0 - g(\theta)'\Sigma^{-1}g(\theta) \leq 0$, the multiplier λ coincides with the derivative of the value function with respect to c_0 and can be interpreted as a marginal *loss* from tightening the admissible tolerance. Under the alternative normalization $g(\theta)'\Sigma^{-1}g(\theta) - c_0 \leq 0$, the same multiplier appears with the opposite sign and is interpreted as a marginal *gain* from relaxing the restriction.

This difference in interpretation carries over to the individual shadow prices, which are proportional to the multiplier λ up to the sign induced by the chosen constraint formulation and are further scaled by a term reflecting the local geometry of each restriction through Σ^{-1}, as discussed below.

Individual shadow prices. The *individual shadow prices* (ISP) are defined as first derivatives of the Lagrangian with respect to individual restriction components, thereby decomposing the total shadow price into restriction-specific marginal effects. To evaluate the effect of relaxing the jth restriction by moving away from the null, we consider a *directional* derivative of the Lagrangian along

$$v_j = \mathrm{sign}(g_j)\, e_j,$$

where e_j denotes the unit vector in the jth coordinate direction. This construction ensures that the derivative is taken in the direction corresponding to relaxation of the restriction, regardless of the sign of g_j.

The two equivalent constraint formulations yield ISP that differ by sign. In particular, when the constraint is written as $(c_0 - g(\theta)'\Sigma^{-1}g(\theta))$, the ISP takes the form

$$\mathrm{ISP}_j := D_{v_j}\mathcal{L}_n = \langle \nabla_g \mathcal{L}_n,\, v_j\rangle = \frac{\partial \mathcal{L}_n}{\partial g_j} \cdot \mathrm{sign}(g_j) = -2\lambda\, \mathrm{sign}(g_j)\, [\Sigma^{-1}g]_j.$$

Under the alternative normalization $(g(\theta)'\Sigma^{-1}g(\theta) - c_0)$, the ISP becomes

$$\mathrm{ISP}_j := D_{v_j}\mathcal{L}_n = \langle \nabla_g \mathcal{L}_n,\, v_j\rangle = \frac{\partial \mathcal{L}_n}{\partial g_j} \cdot \mathrm{sign}(g_j) = 2\lambda\, \mathrm{sign}(g_j)\, [\Sigma^{-1}g]_j.$$

The sign of the ISP depends on the normalization of the constraint and the corresponding economic interpretation of λ. If the constraint is written as $(c_0 - g(\theta)'\Sigma^{-1}g(\theta)) \leq 0$, then λ represents the marginal loss from tightening the admissible tolerance, so that $-\lambda$ corresponds to the marginal gain from relaxing the restriction; in this case, ISP are proportional to $-\lambda$ and therefore appear with negative values. Under the alternative normalization $(g(\theta)'\Sigma^{-1}g(\theta) - c_0) \leq 0$, λ directly captures the marginal gain from relaxing the restriction, and ISP are proportional to λ, yielding positive values.

Importantly, this sign difference is purely a consequence of how the constraint is written and does not alter the economic content of ISP. In all cases, ISP represent the marginal *gain* obtained from relaxing each restriction component. What matters economically is therefore the magnitude of ISP rather than its sign.[1] Thus, the absolute value of ISP measures the size of the marginal improvement in fit associated with relaxing the corresponding restriction. Beyond sign considerations, ISP are further scaled by the local geometry induced by Σ^{-1}, incorporating the researcher's assigned credibility to each restriction and capturing restriction-specific directional sensitivity within the quadratic constraint set.

S.2 Improving inference for validity testing

The ISP-based procedure is not designed to replace validity testing. Rather, it serves as a complementary screening device that identifies empirically relevant restrictions before truth-based inference is carried out. This separation between relevance and validity clarifies the inferential role of ISP and provides the basis for improved power in subsequent validity tests.

S.2.1 Sample splitting and cross-fitted inference for ISP screening and validity testing

The ISP-based test is designed to identify empirically relevant restrictions, whereas the subsequent validity test assesses whether those selected restrictions are satisfied. If both steps are conducted on the same data, the resulting validity tests are subject to post-selection bias. We therefore separate selection and testing by sample splitting, and use cross-fitting to reduce split dependence.

[1] For example, under the first normalization of the constraint, ISP $= -5$ indicates a larger gain from relaxation than ISP $= -3$, even though -3 is numerically greater than -5.

Cross-fitted sample splitting. Partition the sample into $K \geq 2$ folds,

$$\mathcal{I}_1, \dots, \mathcal{I}_K, \qquad \bigcup_{k=1}^{K} \mathcal{I}_k = \{1, \dots, n\}, \qquad \mathcal{I}_k \cap \mathcal{I}_\ell = \varnothing \text{ for } k \neq \ell.$$

Let $\mathcal{I}_{-k} := \{1, \dots, n\} \setminus \mathcal{I}_k$. For each fold k:

1. **Selection step on training sample $\mathcal{I}_{-k}$.** Using only observations in $\mathcal{I}_{-k}$, compute the optimized tolerance

$$\hat{c}^{(-k)} \in \arg\min_{c \in \mathcal{C}_{\text{adm}}} \widehat{R}_{\text{lin}}^{(-k)}(c),$$

obtain the corresponding constrained estimator

$$\hat{\theta}^{(-k)} = \hat{\theta}^{(-k)}(\hat{c}^{(-k)}), \qquad \hat{\lambda}^{(-k)} = \hat{\lambda}^{(-k)}(\hat{c}^{(-k)}),$$

and compute the ISP vector

$$\widehat{ISP}^{(-k)} = 2\hat{\lambda}^{(-k)} \operatorname{diag}\big(\operatorname{sign}(g(\hat{\theta}^{(-k)}))\big) \Sigma^{-1} g(\hat{\theta}^{(-k)}).$$

Apply the plateau rule to $\widehat{ISP}^{(-k)}$, yielding the selected set of empirically relevant restrictions

$$\hat{S}^{(-k)} \subseteq \{1, \dots, q\}.$$

2. **Validity-testing step on held-out sample $\mathcal{I}_k$.** For each $j \in \hat{S}^{(-k)}$, construct a validity test statistic $T_j^{(k)}$ using only observations in $\mathcal{I}_k$. Let $p_j^{(k)}$ denote the corresponding fold-specific p-value.

Because the testing sample $\mathcal{I}_k$ is independent of the training sample $\mathcal{I}_{-k}$, the p-values $p_j^{(k)}$ are valid conditional on the selected set $\hat{S}^{(-k)}$.

Fold-specific validity test. For concreteness, suppose the validity test concerns the null

$$H_{0j} : g_j(\theta^0) = 0.$$

Let $\hat{\theta}^{0(k)}$ denote an unconstrained estimator computed on the held-out sample $\mathcal{I}_k$, and define

$$\hat{g}_j^{(k)} := g_j(\hat{\theta}^{0(k)}).$$

Assume that

$$\sqrt{n_k}\,\hat{g}_j^{(k)} \xrightarrow{d} N\big(0, \sigma_j^2\big), \qquad n_k := |\mathcal{I}_k|,$$

with a consistent variance estimator $\hat{\sigma}_j^{2,(k)}$. Then the fold-specific Wald statistic is

$$T_j^{(k)} = \frac{\sqrt{n_k}\,\hat{g}_j^{(k)}}{\hat{\sigma}_j^{(k)}},$$

and the corresponding two-sided p-value is

$$p_j^{(k)} = 2\Big(1 - \Phi\big(|T_j^{(k)}|\big)\Big). \tag{S.1}$$

Alternatively, $p_j^{(k)}$ may be computed by a bootstrap procedure applied only to the held-out sample $\mathcal{I}_k$.

Assumption S.1 (Cross-fitted fold-wise validity)**.** For each fold k and each $j \in \{1, \ldots, q\}$:

1. conditional on the training sample $\mathcal{I}_{-k}$, the held-out statistic $T_j^{(k)}$ is asymptotically pivotal under H_{0j};
2. the corresponding p-value satisfies

$$\Pr\big(p_j^{(k)} \le \alpha \mid \mathcal{I}_{-k}\big) \le \alpha + o(1) \qquad \text{uniformly in } \alpha \in (0, 1).$$

Theorem S.1 (Conditional validity of fold-specific p-values)**.** Under Assumption S.1, for each fold k and any $j \in \hat{S}^{(-k)}$,

$$\Pr\big(p_j^{(k)} \le \alpha \mid \mathcal{I}_{-k}\big) \le \alpha + o(1).$$

Hence the fold-specific p-values remain valid despite the data-driven ISP selection step.

Proof. Fix a fold k and a restriction index $j \in \hat{S}^{(-k)}$. By construction, the selected set $\hat{S}^{(-k)}$ is determined solely from the training sample $\mathcal{I}_{-k}$, whereas the statistic $T_j^{(k)}$ and its p-value $p_j^{(k)}$ are computed solely from the held-out sample $\mathcal{I}_k$. Since $\mathcal{I}_k$ and $\mathcal{I}_{-k}$ are independent by sample splitting, conditioning on $\mathcal{I}_{-k}$ does not alter the null distribution of the held-out statistic.

Under Assumption S.1, conditional on $\mathcal{I}_{-k}$, the statistic $T_j^{(k)}$ is asymptotically pivotal under H_{0j}, and the corresponding p-value satisfies

$$\Pr\big(p_j^{(k)} \le \alpha \mid \mathcal{I}_{-k}\big) \le \alpha + o(1) \qquad \text{uniformly in } \alpha \in (0,1).$$

Because $j \in \hat{S}^{(-k)}$ is measurable with respect to the training sample $\mathcal{I}_{-k}$, this conditional validity continues to hold after selection. Therefore,

$$\Pr\big(p_j^{(k)} \le \alpha \mid \mathcal{I}_{-k}\big) \le \alpha + o(1) \qquad \text{for each } j \in \hat{S}^{(-k)}.$$

This establishes the result. □

Aggregation across folds. For each restriction j, define the set of folds in which it is selected,

$$\mathcal{K}_j := \{k \in \{1, \dots, K\} : j \in \hat{S}^{(-k)}\}, \qquad m_j := |\mathcal{K}_j|.$$

If $m_j = 0$, the restriction is never selected and no validity test is conducted. If $m_j \ge 1$, collect the valid fold-specific p-values

$$\{p_j^{(k)} : k \in \mathcal{K}_j\}.$$

To obtain a single p-value for H_{0j}, we use the Bonferroni aggregation rule

$$\tilde{p}_j = \min\Big\{1,\ m_j \min_{k \in \mathcal{K}_j} p_j^{(k)}\Big\}. \tag{S.2}$$

This rule is conservative but rigorous and does not require additional dependence restrictions beyond fold-wise validity. More efficient combination rules may be used under stronger assumptions, but (S.2) is sufficient for uniform asymptotic validity.

Theorem S.2 (Validity of aggregated cross-fitted p-values)**.** Under Assumption S.1, for any null restriction H_{0j},

$$\Pr(\tilde{p}_j \le \alpha) \le \alpha + o(1), \qquad \alpha \in (0,1),$$

where $\tilde{p}_j$ is defined in (S.2). Therefore, the aggregated cross-fitted p-value is asymptotically valid for validity testing after ISP-based selection.

Proof. Fix a null restriction H_{0j}. Let

$$\mathcal{K}_j := \{k \in \{1, \dots, K\} : j \in \hat{S}^{(-k)}\}, \qquad m_j := |\mathcal{K}_j|.$$

If $m_j = 0$, then $\tilde{p}_j = 1$ by construction, and hence

$$\Pr(\tilde{p}_j \le \alpha) = 0 \le \alpha + o(1).$$

It therefore suffices to consider the case $m_j \ge 1$.

By definition,

$$\tilde{p}_j = \min\Big\{1,\ m_j \min_{k \in \mathcal{K}_j} p_j^{(k)}\Big\}.$$

Hence the event $\{\tilde{p}_j \le \alpha\}$ implies

$$\min_{k \in \mathcal{K}_j} p_j^{(k)} \le \frac{\alpha}{m_j}.$$

Therefore,

$$\Pr(\tilde{p}_j \le \alpha) \le \Pr\left(\min_{k \in \mathcal{K}_j} p_j^{(k)} \le \frac{\alpha}{m_j}\right). \tag{S.3}$$

Applying the union bound,

$$\Pr\left(\min_{k \in \mathcal{K}_j} p_j^{(k)} \le \frac{\alpha}{m_j}\right) \le \sum_{k \in \mathcal{K}_j} \Pr\left(p_j^{(k)} \le \frac{\alpha}{m_j}\right).$$

For each $k \in \mathcal{K}_j$, Theorem S.1 implies

$$\Pr\left(p_j^{(k)} \le \frac{\alpha}{m_j} \,\middle|\, \mathcal{I}_{-k}\right) \le \frac{\alpha}{m_j} + o(1).$$

Taking expectations and summing over $k \in \mathcal{K}_j$ yields

$$\sum_{k \in \mathcal{K}_j} \Pr\left(p_j^{(k)} \le \frac{\alpha}{m_j}\right) \le m_j \cdot \frac{\alpha}{m_j} + o(1) = \alpha + o(1).$$

Combining this with (S.3), we obtain

$$\Pr(\tilde{p}_j \le \alpha) \le \alpha + o(1).$$

This proves asymptotic validity of the aggregated cross-fitted p-value. □

S.2.2 Why ISP-based screening can improve the power of subsequent validity tests

The purpose of the ISP-based test is not to replace validity testing, but to concentrate validity testing on restrictions that are empirically relevant for model fit. This concentration can improve power by reducing the multiplicity burden and eliminating low-information restrictions from the testing family.

Consider a family of null hypotheses

$$\mathcal{H} = \{H_{0j} : g_j(\theta^0) = 0,\ j = 1, \dots, q\}.$$

Suppose the researcher wishes to control familywise error rate at level α. Under Bonferroni correction, testing all q restrictions uses the per-hypothesis threshold

$$\alpha_q = \frac{\alpha}{q}.$$

Now let $\hat{S} \subseteq \{1, \dots, q\}$ be the set of empirically relevant restrictions selected by the ISP procedure, with cardinality $|\hat{S}| = m$. Conditional on a given selected set S, the corresponding Bonferroni threshold becomes

$$\alpha_m = \frac{\alpha}{m}.$$

Since $m \leq q$, it follows that $\alpha_m \geq \alpha_q$. Thus, whenever the ISP procedure removes empirically negligible restrictions from the testing family, the critical value for the remaining validity tests becomes less stringent.

To formalize the power gain, let $j \in S$ be a false null with fold-specific p-value p_j. Define the power functions

$$\pi_q(j) := \Pr\left(p_j \leq \frac{\alpha}{q}\right), \qquad \pi_m(j) := \Pr\left(p_j \leq \frac{\alpha}{m}\right).$$

Because $\alpha/m \geq \alpha/q$, monotonicity of rejection probabilities implies

$$\pi_m(j) \geq \pi_q(j). \tag{S.4}$$

Proposition S.1 (Conditional power improvement from ISP-based screening)**.** Fix a selected set S with cardinality m, and suppose that $j \in S$ is a false null. If familywise error is controlled by Bonferroni correction, then the validity test conducted only on S has weakly greater power for testing H_{0j} than the corresponding test conducted on the full family $\mathcal{H}$. In particular,

$$\Pr\left(p_j \le \frac{\alpha}{m} \,\middle|\, \hat{S} = S\right) \ge \Pr\left(p_j \le \frac{\alpha}{q} \,\middle|\, \hat{S} = S\right).$$

The inequality is strict whenever

$$\Pr\left(\frac{\alpha}{q} < p_j \le \frac{\alpha}{m} \,\middle|\, \hat{S} = S\right) > 0.$$

Proof. Fix a selected set $S \subseteq \{1, \ldots, q\}$ with cardinality m, and suppose $j \in S$ is a false null. Under Bonferroni correction, testing over the full family $\mathcal{H}$ uses threshold α/q, while testing only over the selected family S uses threshold α/m. Since $m \le q$, we have

$$\frac{\alpha}{m} \ge \frac{\alpha}{q}.$$

Now consider the event inclusion

$$\left\{p_j \le \frac{\alpha}{q}\right\} \subseteq \left\{p_j \le \frac{\alpha}{m}\right\}.$$

Taking conditional probabilities given $\hat{S} = S$ yields

$$\Pr\left(p_j \le \frac{\alpha}{m} \,\middle|\, \hat{S} = S\right) \ge \Pr\left(p_j \le \frac{\alpha}{q} \,\middle|\, \hat{S} = S\right).$$

This proves weak power improvement.

Moreover, the inequality is strict whenever the set difference has positive probability, that is, whenever

$$\Pr\left(\frac{\alpha}{q} < p_j \le \frac{\alpha}{m} \,\middle|\, \hat{S} = S\right) > 0.$$

In that case,

$$\Pr\left(p_j \le \frac{\alpha}{m} \,\middle|\, \hat{S} = S\right) > \Pr\left(p_j \le \frac{\alpha}{q} \,\middle|\, \hat{S} = S\right).$$

This establishes the proposition. □

Remark S.1 (Interpretation)**.** The power improvement is conditional rather than uniform. Screening helps when the ISP step successfully removes low-information restrictions that would otherwise inflate the multiplicity adjustment. However, if a substantively false restriction is screened out because its ISP is too small, then it will not be tested in the second step. Thus, the gain in power is targeted toward empirically relevant restrictions rather than global detection of all possible violations.

Remark S.2 (Connection to informativeness)**.** The result above clarifies the role of the ISP-based test. ISP does not test whether a restriction is true. Instead, it tests whether model fit is sensitive to that restriction. By concentrating validity testing on the restrictions to which fit is most sensitive, the procedure improves the allocation of inferential power.

Remark S.3 (Why sample splitting is needed)**.** Without sample splitting, the same sample would be used both to identify large ISPs and to test the corresponding restrictions. This conditions the testing step on extreme selection events and generally invalidates standard p-values. Cross-fitting restores validity by ensuring that each fold-specific validity test is conducted on data independent of the ISP-based screening step that selected the restriction.

Remark S.4 (Bootstrap within cross-fitting)**.** When sample splitting or cross-fitting is used, the bootstrap-based plateau rule is implemented within each training fold $\mathcal{I}_{-k}$. That is, for each k, the bootstrap recomputes the optimized tolerance, the constrained estimator, and the ISP ranking using only the training observations. The resulting selected sets $\hat{S}^{(-k)}$ are then carried to the held-out fold $\mathcal{I}_k$ for validity testing. This preserves the separation between selection and inference.